

Shezo

1. Basic Integration Rules

1. Constant Rule:

$$\int c \, dx = cx + C$$

1. Power Rule:

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

1. Constant Multiple Rule:

$$\int c f(x) \, dx = c \int f(x) \, dx$$

1. Sum and Difference Rule:

$$\int [f(x) \pm g(x)] \, dx = \int f(x) \, dx \pm \int g(x) \, dx$$

2. Integration of Exponential and Logarithmic Functions

1. Exponential Functions:

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C, \\ \text{quad } a > 0, a \neq 1]$$

1. Logarithmic Function:

$$\int \frac{1}{x} dx = \ln |x| + C$$

3. Integration of Trigonometric Functions

1. Basic Trigonometric Functions:

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C$$

$$\int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

$$\int \csc x \, dx = \ln |\csc x - \cot x| + C$$

1. Inverse Trigonometric Functions:

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$\int \frac{-1}{\sqrt{1-x^2}} dx = \cos^{-1} x + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$$

$$\int \frac{-1}{1+x^2} dx = \cot^{-1} x + C$$

$$\int \frac{1}{|x|\sqrt{x^2-1}} dx = \sec^{-1} x + C$$

$$\int \frac{-1}{|x|\sqrt{x^2-1}} dx = \csc^{-1} x + C$$

4. Integration by Parts

$$\int uv \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$$

5. Integration of Rational Functions

1. Integration of $\frac{1}{a^2+x^2}$:

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

1. Integration of $\frac{1}{\sqrt{a^2-x^2}}$:

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$$

1. Integration of $\frac{1}{x^2-a^2}$:

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

6. Integration of Hyperbolic Functions

1. Basic Hyperbolic Functions:

$$\int \sinh x \, dx = \cosh x + C$$

$$\int \cosh x \, dx = \sinh x + C$$

$$\int \tanh x \, dx = \ln |\cosh x| + C$$

$$\int \coth x \, dx = \ln |\sinh x| + C$$

1. Inverse Hyperbolic Functions:

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \sinh^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \left(\frac{x}{a} \right) + C$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

7. Special Integrals

1. Gaussian Integral (Non-elementary):

$$\int e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \text{ (error function, erf(x))}$$

1. Dirac Delta Function:

$$\int_{-\infty}^{\infty} \delta(x) f(x) dx = f(0)$$

1. Beta Function:

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

1. Gamma Function:

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx$$

1. Basic Limit Rules

1. Constant Rule:

$$\lim_{x \rightarrow a} c = c$$

1. Identity Rule:

$$\lim_{x \rightarrow a} x = a$$

1. Sum and Difference Rule:

$$\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

1. Product Rule:

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

1. Quotient Rule (if denominator $\neq 0$):

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

1. Power Rule:

$$\lim_{x \rightarrow a} [f(x)]^n = (\lim_{x \rightarrow a} f(x))^n$$

2. Important Standard Limits

1. Limit of a Constant Function:

$$\lim_{x \rightarrow a} c = c$$

1. Limit of a Polynomial Function:

$$\lim_{x \rightarrow a} P(x) = P(a)$$

1. Limit of a Rational Function:

$$\lim_{x \rightarrow a} \frac{P(x)}{Q(x)} = \frac{P(a)}{Q(a)}$$

1. Limits Involving Infinity:

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0, \quad n > 0$$

3. Trigonometric Limits

1. Fundamental Limit:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

1. Cosine Limit:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

1. Tan Limit:

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

4. Exponential and Logarithmic Limits

1. Exponential Limit:

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

1. Logarithmic Limit:

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

5. L'Hôpital's Rule (Indeterminate Forms: $\frac{0}{0}$ or $\frac{\infty}{\infty}$)

If $\lim_{x \rightarrow a} f(x) = 0$ **and**
 $\lim_{x \rightarrow a} g(x) = 0$ **, then:**

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

6. Special Indeterminate Limits

1. Infinity over Infinity:

$$\lim_{x \rightarrow \infty} \frac{ax^m}{bx^n} = \begin{cases} 0, & \text{if } m < n \\ \frac{a}{b}, & \text{if } m = n \\ \infty, & \text{if } m > n \end{cases}$$

1. Power Form ($0^0, \infty^0, 1^\infty$):

Convert to logarithmic form before evaluating.

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Differentiation :-

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$$1. \frac{d}{dx}(c) = 0$$

$$2. \frac{d}{dx}(x^n) = nx^{n-1}$$

$$3. \frac{d}{dx}(c \cdot f(x)) = c \cdot f'(x)$$

~~4.~~

$$4. \frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$$

$$5. \frac{d}{dx}(e^x) = e^x$$

$$6. \frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$7. \frac{d}{dx}(\sin x) = \cos x$$

$$8. \frac{d}{dx}(\cos x) = -\sin x$$

$$9. \frac{d}{dx}(\tan x) = \sec^2 x$$

$$10. \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$11. \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$12. \frac{d}{dx}(\cot x) = -\operatorname{csc}^2 x$$

$$13. \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\left(\frac{1}{x} \right) \left(x \sqrt{1-x^2} - \int \sin^{-1}(u) x \right)$$

14.

$$\frac{d}{dx} (\cos^{-1} x) = - \frac{1}{\sqrt{1-x^2}}$$

$$15. \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$$

$$16. \frac{d}{dx} (\cot^{-1} x) = - \frac{1}{1+x^2}$$

$$17. \frac{d}{dx} (\sec^{-1} x) = \frac{1}{|x| \sqrt{x^2-1}}$$

$$18. \frac{d}{dx} (\csc^{-1} x) = - \frac{1}{|x| \sqrt{x^2-1}}$$

19. ~~Product~~ rule $\leftarrow$ Quotient

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$$

20. Product rule

$$\frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

21. Chain Rule

$$\frac{d}{dx} \quad \text{if} \quad y = f(g(x))$$

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

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22. 2nd derivative

$$f''(x) = \frac{d^2 y}{dx^2}$$

23.

nth Derivative

$$f^{(n)}(x) = \frac{d^n y}{dx^n}$$